Team Clicks and Conflicts

Do Highly Engaged Youtube Videos Have More Extreme Comments?

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Background

YouTube

- With 2.5B+ monthly users, YouTube's comment sections is a major site of social interaction
- Comments are typically treated as engagement metrics, not as indicators of discourse quality
- Gap: We lack a clear understanding of how video-level features relate to comment environments:
    - Titles, categories, publish dates
    - Popularity signals (views, likes, comment counts)

Our Approach

- Built an NLP-based pipeline to:
    - Link comments to video metadata
    - Quantify sentiment and toxicity/extremeness

Significance

Importance for Social Computing

- Comment sections are real-time social systems, not just metrics
- Engagement signals are not neutral
- Potential feedback loop:
    - Extreme content → more engagement → more visibility → more extreme discourse

Broader Impact

- Insights for platform design & governance
- Extends beyond YouTube:
    - Helps explain how attention-driven platforms shape online hostility, especially in news and political discourse

Related Works

Paper	Study/Limitations	Our Study's Advancements
Amin et al. Sentiment Analysis On YouTube Comments Using Machine Learning Techniques Based On Video Games Content	- Relied on TextBlob → limited ability to read nuance/complexity in comments - Limited to video game content	- Used VADER and Detoxify for precise extremeness analysis - Generalized research question by performing cross-categorical analysis.
Beknazar-Yuzbashev et al. Toxic Content and User Engagement on Social Media	- Found that toxic content drives engagement (on Facebook, Twitter, YouTube)	- Investigate inverse relationship and deeper look at engagement (views, likes, comments). - Our null result challenges the paper's assumption.
Lorentz et al. Sentiment analysis on YouTube comments to predict YouTube video like proportions	- Found inconclusive correlation between sentiment and likes. - Sampling bias that resulted in inflated correlation predictions.	- Corrected bias by using dataset with greater variation in engagement.
Mili et al. Engagement, User Satisfaction, and the Amplification of Divisive content on Social Media	- Argue that YouTube algorithm amplifies extreme posts	- Our null correlation challenges the paper's argument.
Dehkhoda et al. Analyzing toxicity in YouTube comments with the help of Machine Learning	- Analysed sentiment-extremity correlation in comments using small dataset (~3700)	- Scaled extremity-engagement analysis to >600,000 data points.

Research Questions/Goals/Objectives

Research Question: Do YouTube videos with higher engagement receive more intense or extreme comments?

Our goals:

- Classify comments as strongly positive or strongly negative.
- Measure video engagements considering like, views, and number of comments.
- Compare a video's comments to its engagement
- Identify patterns across video categories
- Manually review a sample of comments

Our outcomes:

- Insights on engagement-extremeness correlation
- Data visuals representing correlation
- Clean, merged dataset containing video/comment data

Data and Rationale



Data Source

- Kaggle Youtube Comments Dataset
- Contains **691,350** comments
- With **1,968** unique video IDs



Why?

- Comments capture **content** + **intensity** of expression
- YouTube = high interaction platform

Data Processing

Data Preparation

- Dropped missing comments
- Cleaned text
- Matched video IDs using API

Data Enrichment

- Added engagement metrics
  - Views, likes, comment counts

Challenges

- Large dataset -> batching + sampling
- Missing API data -> filtered

Approach

Step 1: Text

Youtube Comments From Kaggle

Step 2: Features

Sentiment Score
(VADER)
[-1, 1]
Negative -> Positive

Toxicity Score
(Detoxify)
[0,1]
Low -> High

Step 3: Aggregation

Extremeness Score

Extremeness = (|sentiment| + toxicity) / 2

Aggregate by video_id
Avg_extremeness = mean(extremeness)

Step 4: Analysis

Engagement Score

Log + normalization of views, likes, comments

Approach

Strengths

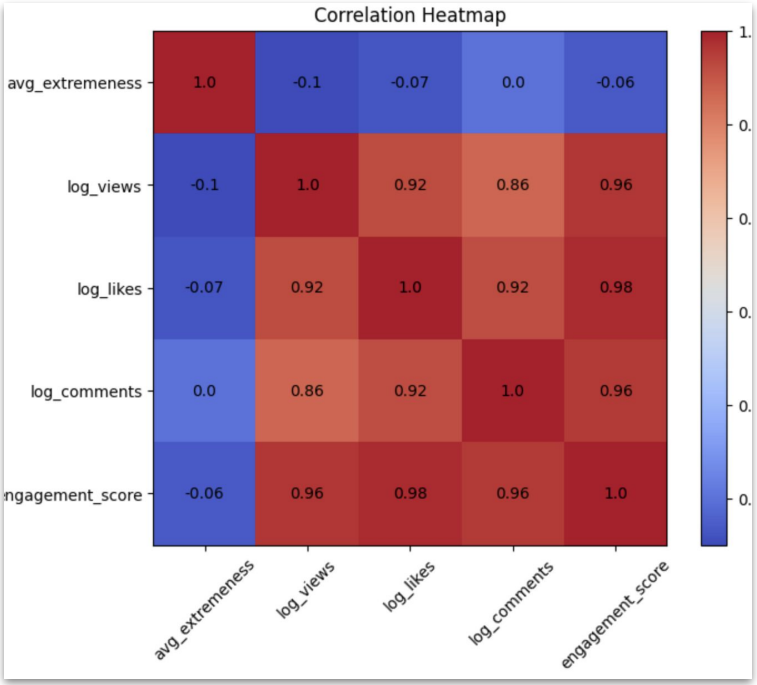
- Consistent Measurement
- Integrates Multiple Signals
- Detects Large Scale Patterns
- Generalizable Pipeline

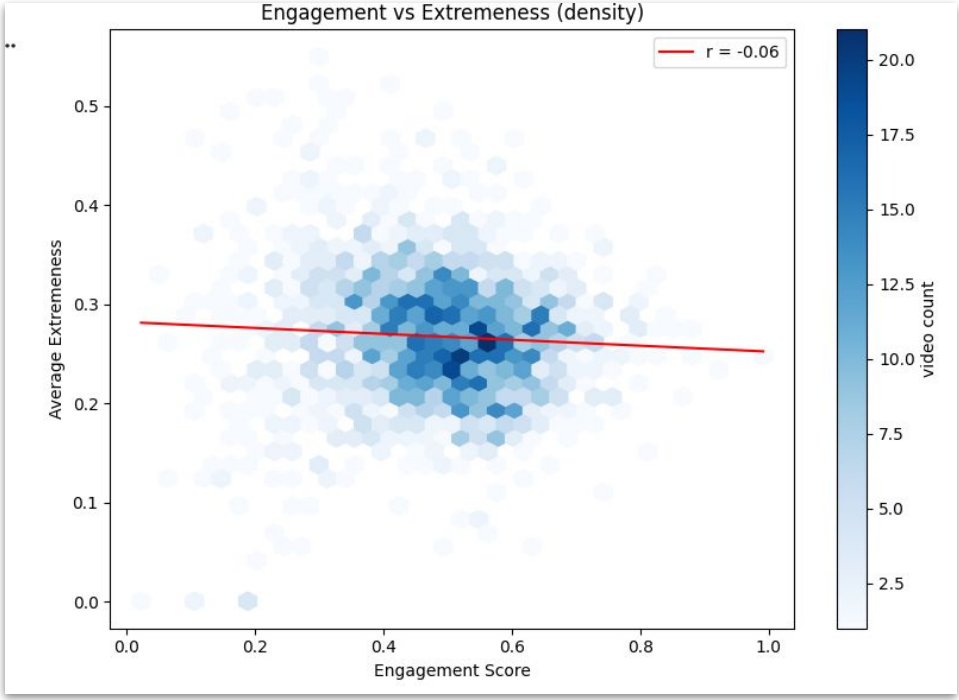
Limitations

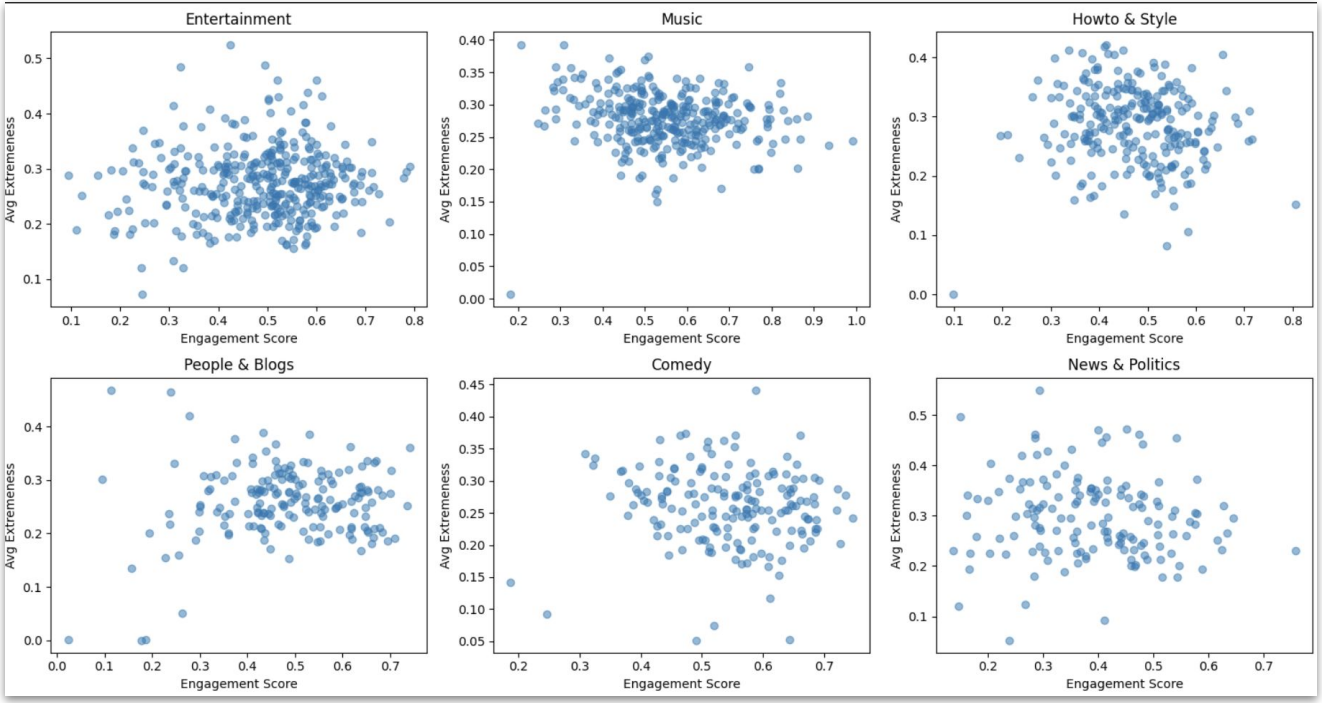
- Extremeness is a Proxy Measure
- Aggregation Hides Variation
- Youtube Commenter Bias

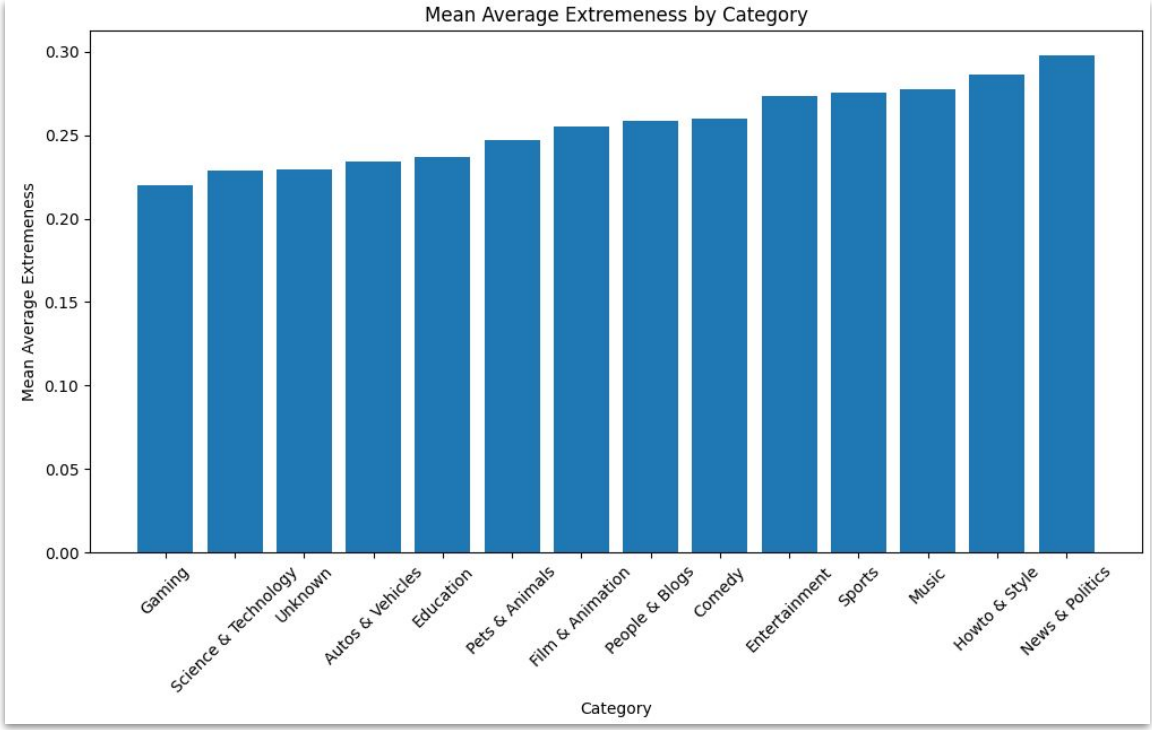
Summary

- We analyzed **1,968 unique YouTube videos** and **691,350 comments**, paired with their associated comment-level extremeness scores
- Our primary finding was that **they are not related**, or at least not in the direction predicted by prior work on engagement-driven amplification across pooled video level and video level by category
    - Across the full sample, the Pearson correlation between composite engagement score and average comment extremeness is **r = -0.06** (R^2 = 0.00)
- We did see correlation between extremeness and category type


Correlation Heatmap
avg_extremeness
log_views
log_likes
log_comments
engagement_score
1.0 -0.1 -0.07 0.0 -0.06
-0.1 1.0 0.92 0.86 0.96
-0.07 0.92 1.0 0.92 0.98
0.0 0.86 0.92 1.0 0.96
-0.06 0.96 0.98 0.96 1.0


Engagement vs Extremeness (density)
r = -0.06
Average Extremeness
Engagement Score
video count
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.8
1.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Entertainment
Music
Howto & Style
People & Blogs
Comedy
News & Politics
Avg Extremeness
Engagement Score


Mean Average Extremeness by Category
Mean Average Extremeness
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Gaming
Science & Technology
Unknown
Autos & Vehicles
Education
Pets & Animals
Film & Animation
People & Blogs
Comedy
Entertainment
Sports
Music
Howto & Style
News & Politics
Category

Implications/Learned/Conclusion

1. Engagement does not predict comment extremeness
2. Within categories, popular videos tend to draw less extreme commentary, not more
3. Content category is the strongest predictor of comment extremeness
4. Platforms likely shape discourse through topic selection, not engagement amplification
5. Future work should examine comment-thread dynamic, where reply behavior may differ from video-level patterns

Thank You!
Any Questions?